

100% Free Implementation Plan: Autonomous Financial Research Agent

1. Prerequisites and Knowledge Requirements

To successfully build this project, you should have a solid foundation in Python and a conceptual understanding of Large Language Models (LLMs). Since you are already learning RAG and LangChain, this project will serve as the perfect practical application of those skills.

Core Knowledge Areas

Domain	Key Concepts to Master
Python Programming	Asynchronous programming (<code>asyncio</code>), environment variables (<code>python-dotenv</code>), and basic data structures.
LangChain Framework	Agents, Tools, Chains, and Memory management. Specifically, focus on <code>LangGraph</code> for more complex agentic workflows.
RAG (Retrieval-Augmented Generation)	Document loading, text splitting, embedding generation, and vector database querying.
API Integration	Handling JSON responses, managing rate limits, and secure API key storage.

2. Development Environment Setup

We will use a standard Python environment. For a student, the most efficient way to manage this is through a local code editor or a cloud-based environment that offers a free tier.

- **Programming Language:** Python 3.9+
- **Editor:** Visual Studio Code (VS Code) or Cursor (which has a great free tier for AI-assisted coding).
- **Package Management:** `pip` or `poetry` .
- **Version Control:** GitHub (Essential for your resume).

3. 100% Free API and Tool Catalog

The following resources are selected specifically because they offer "Free Forever" tiers or generous trial credits without requiring a credit card.

Resource Type	Recommended Tool	Why It's Perfect for Students
LLM API	Groq (Llama 3)	Extremely fast inference with a generous free tier. No credit card required for basic usage.
Search API	Serper.dev	Provides 2,500 free search queries upon signup. Excellent for gathering real-time news.
Financial Data	yfinance (Library)	A Python library that scrapes Yahoo Finance. 100% free, no API key or credit card needed.
Vector Database	ChromaDB	An open-source, local vector database. Runs entirely on your machine for free.
UI Framework	Streamlit	The industry standard for building AI demos quickly. Open-source and free to use.

4. Step-by-Step Implementation Procedure

This roadmap breaks the project down into manageable phases, ensuring you can explain each part to an interviewer.

Phase 1: Foundation and API Integration

- Set up your environment:** Create a GitHub repository and initialize a Python virtual environment.
- API Key Management:** Sign up for Groq and Serper.dev. Store your keys in a `.env` file (never commit this to GitHub).
- Basic Tool Creation:** Write Python functions to fetch stock data using `yfinance` and search for news using Serper.

Phase 2: Building the RAG Pipeline

1. **Document Loading:** Implement a logic to take the search results and "scrape" the text content.
2. **Embedding & Storage:** Use a free embedding model (like `sentence-transformers` which runs locally) to convert text to vectors and store them in ChromaDB.
3. **Retrieval Logic:** Create a function that takes a query, searches ChromaDB, and returns the most relevant text snippets.

Phase 3: Agent Orchestration with LangChain

1. **Define the Agent:** Use LangChain to create an agent that has access to your `yfinance` tool and your `Search/RAG` tool.
2. **Prompt Engineering:** Design a "System Prompt" that instructs the agent to act as a professional financial analyst.
3. **Memory Implementation:** Add `ConversationBufferMemory` so the agent remembers previous parts of the research.

Phase 4: User Interface and Deployment

1. **Build the UI:** Use Streamlit to create a simple text input for the company name and a display area for the final report.
2. **Testing:** Run the agent for several companies (e.g., Apple, Tesla, Nvidia) and refine the prompts based on the output quality.
3. **Documentation:** Write a comprehensive `README.md` for your GitHub repository, explaining how to run the project.

5. Sources to Visit for Learning

To deepen your understanding while building, I recommend the following 100% free resources:

- **LangChain Documentation:** The official [LangChain Conceptual Guide](#) is the best place to understand agents.
- **DeepLearning.ai:** Look for their "Short Courses" on LangChain and RAG. Many are free for a limited time or have free audit options.
- **YouTube - Krish Naik / Greg Kamradt:** Both provide excellent, free tutorials on building end-to-end LLM applications.
- **Streamlit Gallery:** Visit the [Streamlit Gallery](#) for inspiration on how to design your agent's dashboard.

6. Project Roadmap (Timeline)

- **Day 1-2:** Learn basic LangChain Tool usage and integrate `yfinance` .
- **Day 3-5:** Build the RAG pipeline with ChromaDB and Serper.dev.
- **Day 6-8:** Orchestrate the Agent and refine the System Prompt.
- **Day 9-10:** Build the Streamlit UI and document the project on GitHub.